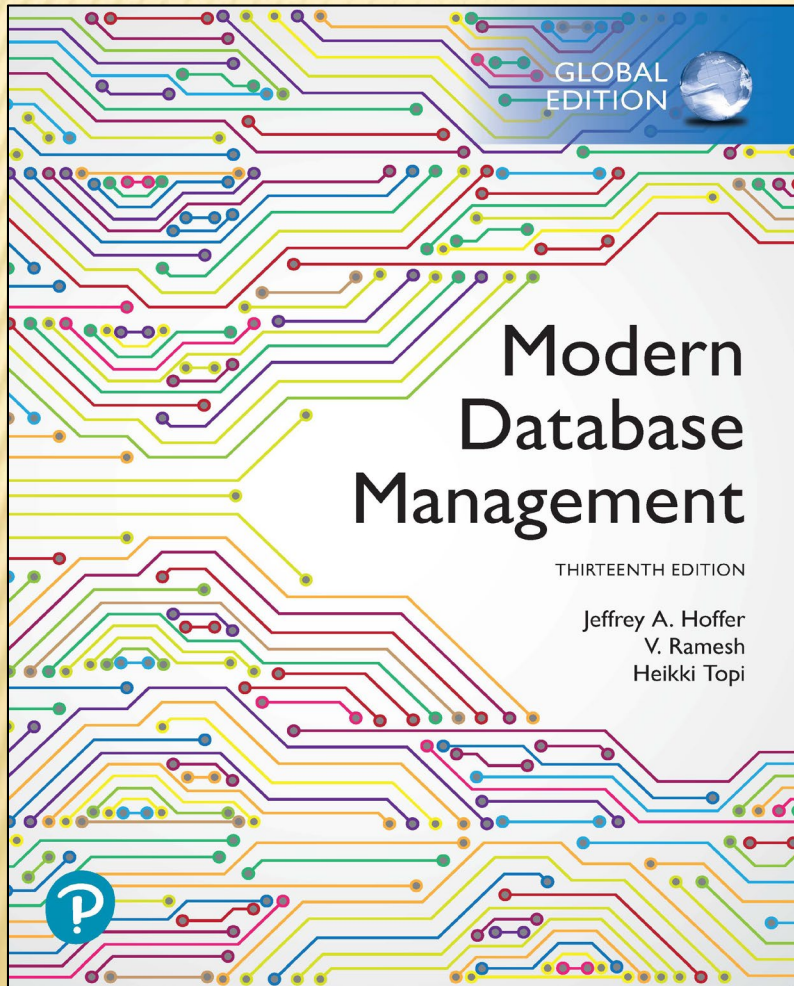


Modern Database Management

Thirteenth Edition



Chapter 9

Data Warehousing

OBJECTIVES

- ✖ Define terms
- ✖ Give reasons for information gap between information needs and availability
- ✖ List reasons for need of data warehousing
- ✖ Describe three levels of data warehouse architectures
- ✖ Describe two components of star schema
- ✖ Estimate fact table size
- ✖ Design a data mart
- ✖ Develop requirements for a data mart
- ✖ Understand future data warehousing trends

DEFINITIONS

× Data Warehouse

- + A **subject-oriented, integrated, time-variant, non-updatable** collection of data used in support of managing decision-making processes
 - × ***Subject-oriented:*** e.g. customers, patients, students, products
 - × ***Integrated:*** consistent naming conventions, formats, encoding structures; from multiple data sources
 - × ***Time-variant:*** can study trends and changes
 - × ***Non-updatable:*** read-only, periodically refreshed

× Data Mart

- + A data warehouse that is limited in scope

NEED FOR DATA WAREHOUSING

- ✖ Integrated, company-wide view of high-quality information (from disparate databases)
- ✖ Separation of *operational* and *informational* systems and data (for improved performance)

ISSUES WITH COMPANY-WIDE VIEW

- ✗ Inconsistent key structures
- ✗ Synonyms
- ✗ Free-form vs. structured fields
- ✗ Inconsistent data values
- ✗ Missing data

See figure 9-1 for example

Figure 9-1
Examples of
heterogeneous
data

STUDENT DATA

<u>StudentNo</u>	LastName	MI	FirstName	Telephone	Status	...
123-45-6789	Enright	T	Mark	483-1967	Soph	
389-21-4062	Smith	R	Elaine	283-4195	Jr	

STUDENT EMPLOYEE

<u>StudentID</u>	Address	Dept	Hours	...
123-45-6789	1218 Elk Drive, Phoenix, AZ 91304	Soc	8	
389-21-4062	134 Mesa Road, Tempe, AZ 90142	Math	10	

STUDENT HEALTH

<u>StudentName</u>	Telephone	Insurance	ID	...
Mark T. Enright	483-1967	Blue Cross	123-45-6789	
Elaine R. Smith	555-7828	?	389-21-4062	

ORGANIZATIONAL TRENDS MOTIVATING DATA WAREHOUSES

- ✗ No single system of records
- ✗ Multiple systems not synchronized
- ✗ Organizational need to analyze activities in a balanced way
- ✗ Customer relationship management
- ✗ Supplier relationship management

SEPARATING OPERATIONAL AND INFORMATIONAL SYSTEMS

- ✗ **Operational system** – a system that is used to run a business in real time, based on current data; also called a system of record
- ✗ **Informational system** – a system designed to support decision making based on historical point-in-time and prediction data for complex queries or data-mining applications

TABLE 9-1 Comparison of Operational and Informational Systems

Characteristic	Operational Systems	Informational Systems
Primary purpose	Run the business on a current basis	Support managerial decision making
Type of data	Current representation of state of the business	Historical point-in-time (snapshots) and predictions
Primary users	Clerks, salespersons, administrators	Managers, business analysts, customers
Scope of usage	Narrow, planned, and simple updates and queries	Broad, ad hoc, complex queries and analysis
Design goal	Performance: throughput, availability	Ease of flexible access and use
Volume	Many constant updates and queries on one or a few table rows	Periodic batch updates and queries requiring many or all rows

DATA WAREHOUSE ARCHITECTURES

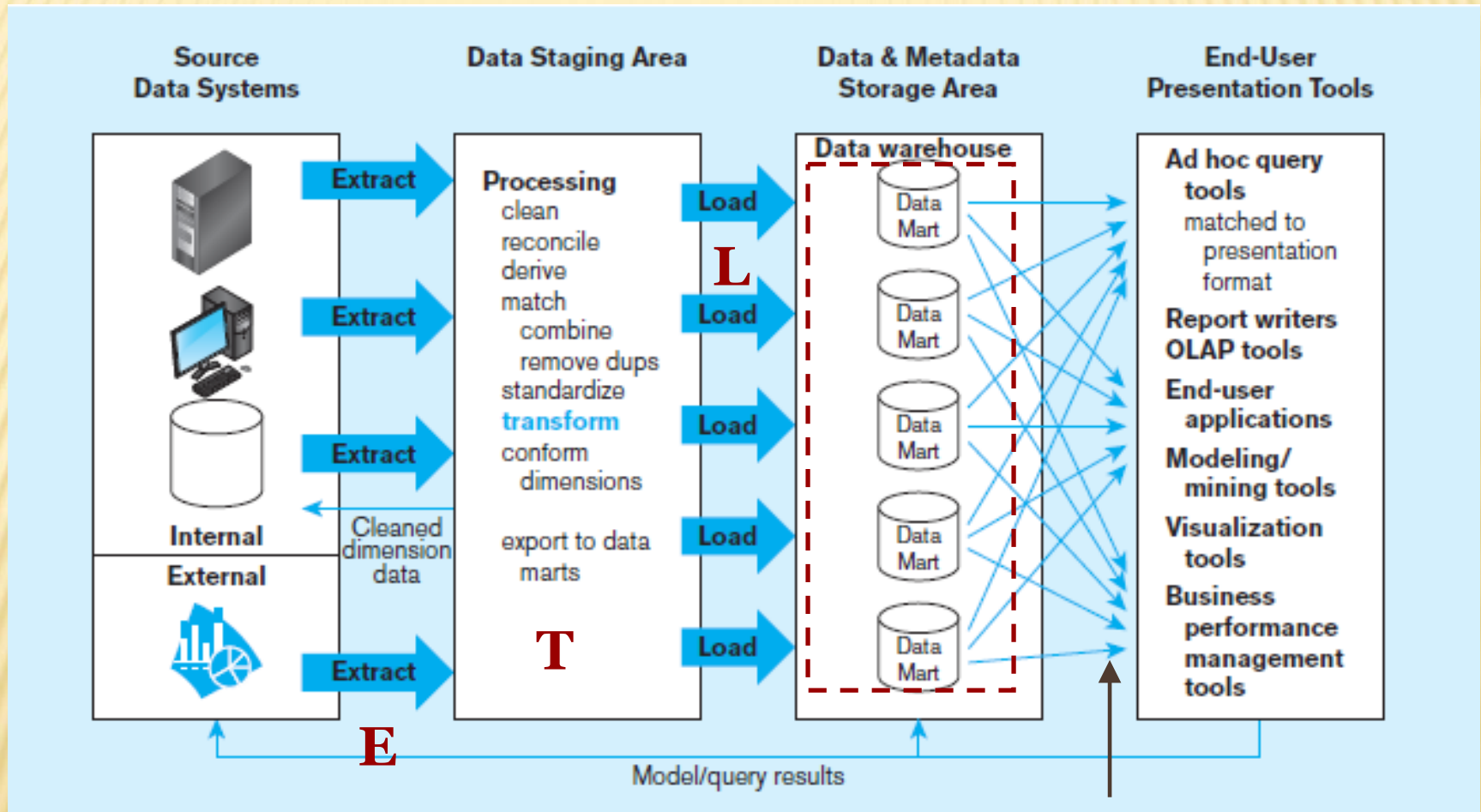
- ✗ Independent Data Mart
- ✗ Dependent Data Mart and Operational Data Store
- ✗ Logical Data Mart and Real-Time Data Warehouse
- ✗ Three-Layer architecture

All involve some form of *extract, transform and load (ETL)*

Figure 9-2 Independent data mart data warehousing architecture

Data marts:

Mini-warehouses, limited in scope



Separate ETL for each *independent* data mart

Data access complexity due to *multiple* data marts

LIMITATIONS OF INDEPENDENT DATA MARTS

- ✖ Separate ETL process for each data mart → redundant data and processing
- ✖ Inconsistency between data marts
- ✖ Difficult to drill down for related facts between data marts
- ✖ Excessive scaling costs as more applications are built
- ✖ High cost for obtaining consistency between marts

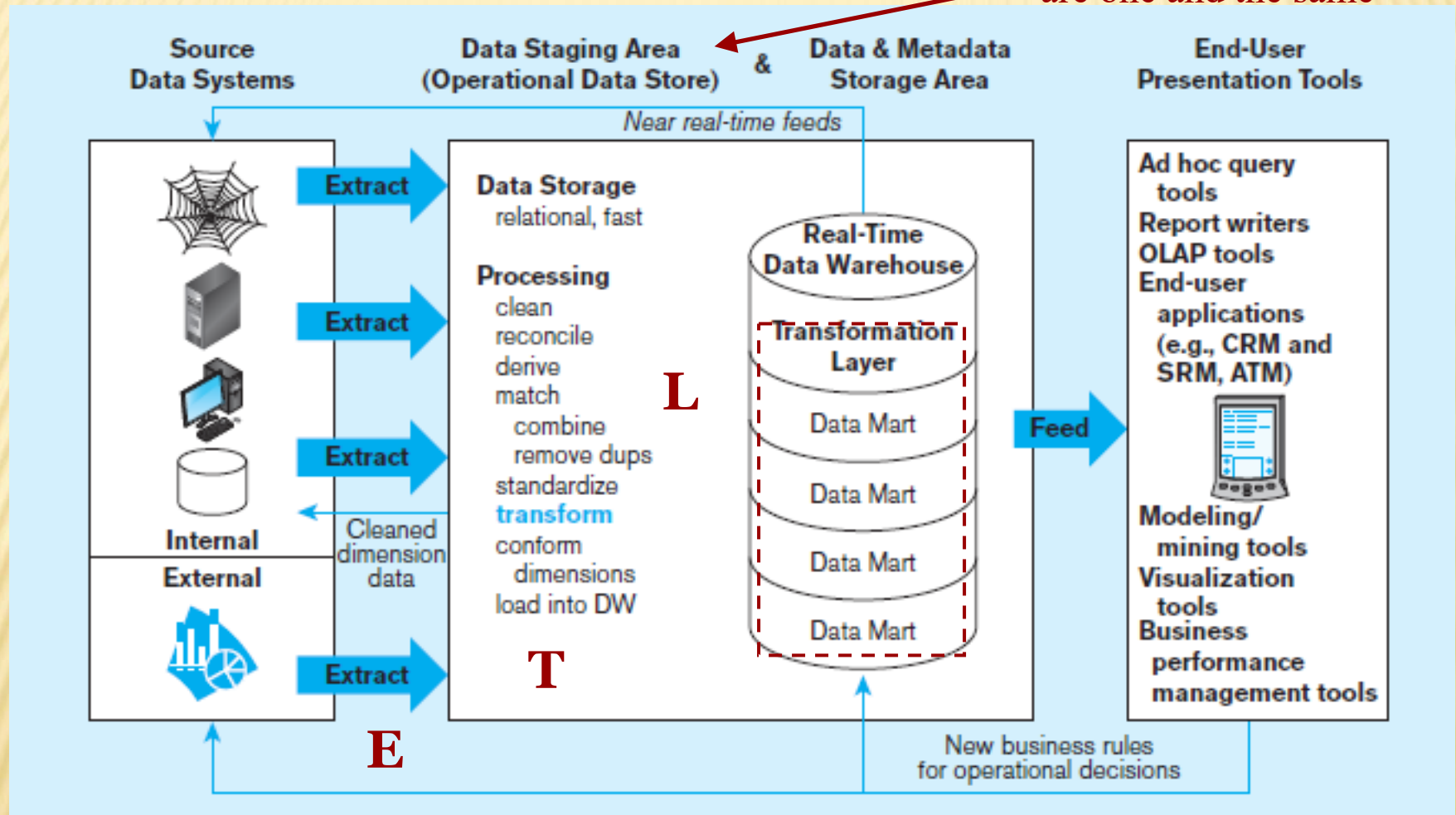
ODS provides option for obtaining *current* data



Simpler data access

Figure 9-4 Logical data mart and real time warehouse architecture

ODS and **data warehouse** are one and the same



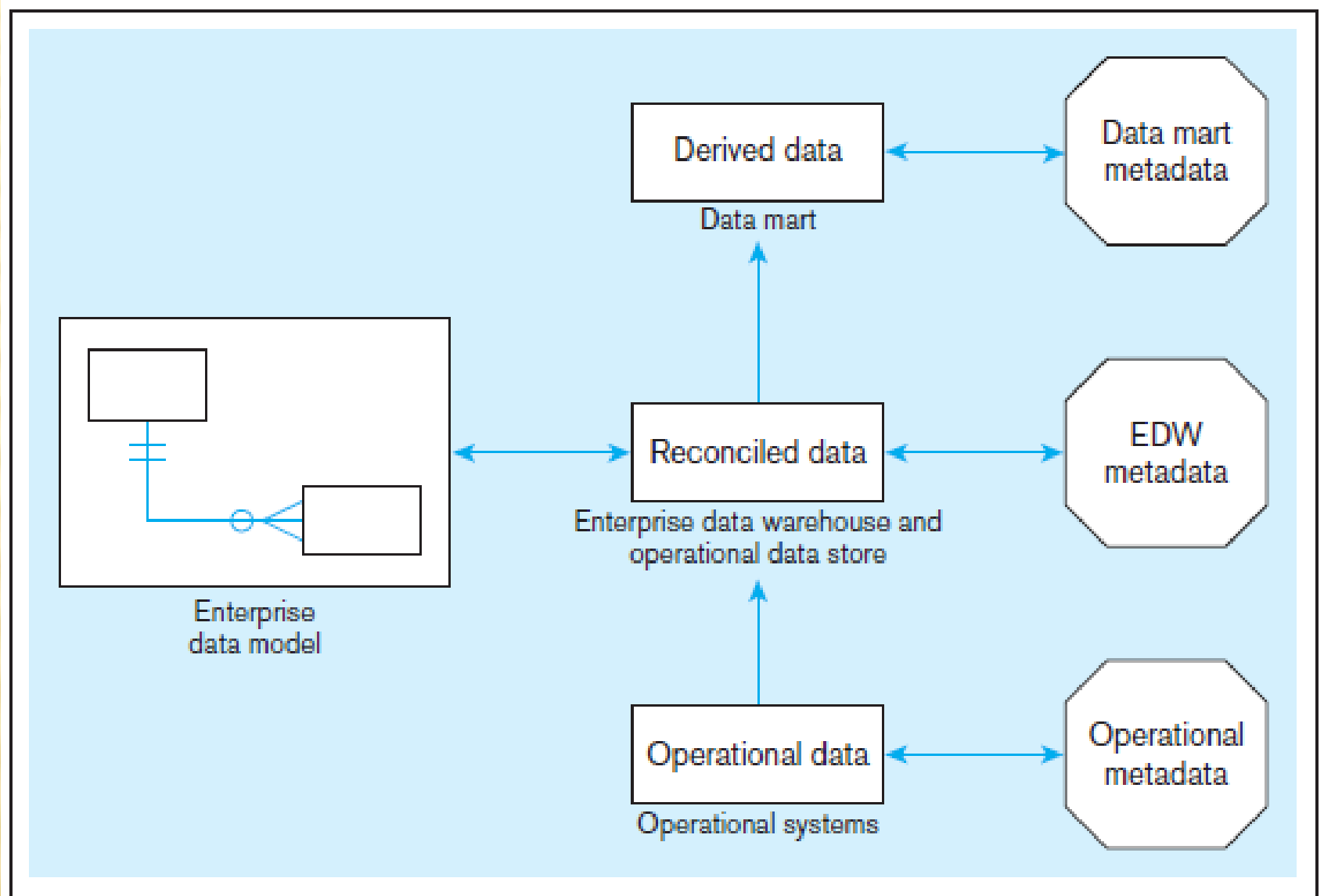
Near real-time ETL for
Data Warehouse

Data marts are NOT separate databases,
but logical *views* of the data warehouse
➔ Easier to create new data marts

TABLE 9-2 Data Warehouse Versus Data Mart

Data Warehouse	Data Mart
Scope • Application independent • Centralized, possibly enterprise-wide • Planned	Scope • Specific DSS application • Decentralized by user area • Organic, possibly not planned
Data • Historical, detailed, and summarized • Lightly denormalized	Data • Some history, detailed, and summarized • Highly denormalized
Subjects • Multiple subjects	Subjects • One central subject of concern to users
Sources • Many internal and external sources	Sources • Few internal and external sources
Other Characteristics • Flexible • Data oriented • Long life • Large • Single complex structure	Other Characteristics • Restrictive • Project oriented • Short life • Starts small, becomes large • Multi, semi-complex structures, together complex

Figure 9-5 Three-layer data architecture for a data warehouse



DATA CHARACTERISTICS

STATUS VS. EVENT DATA

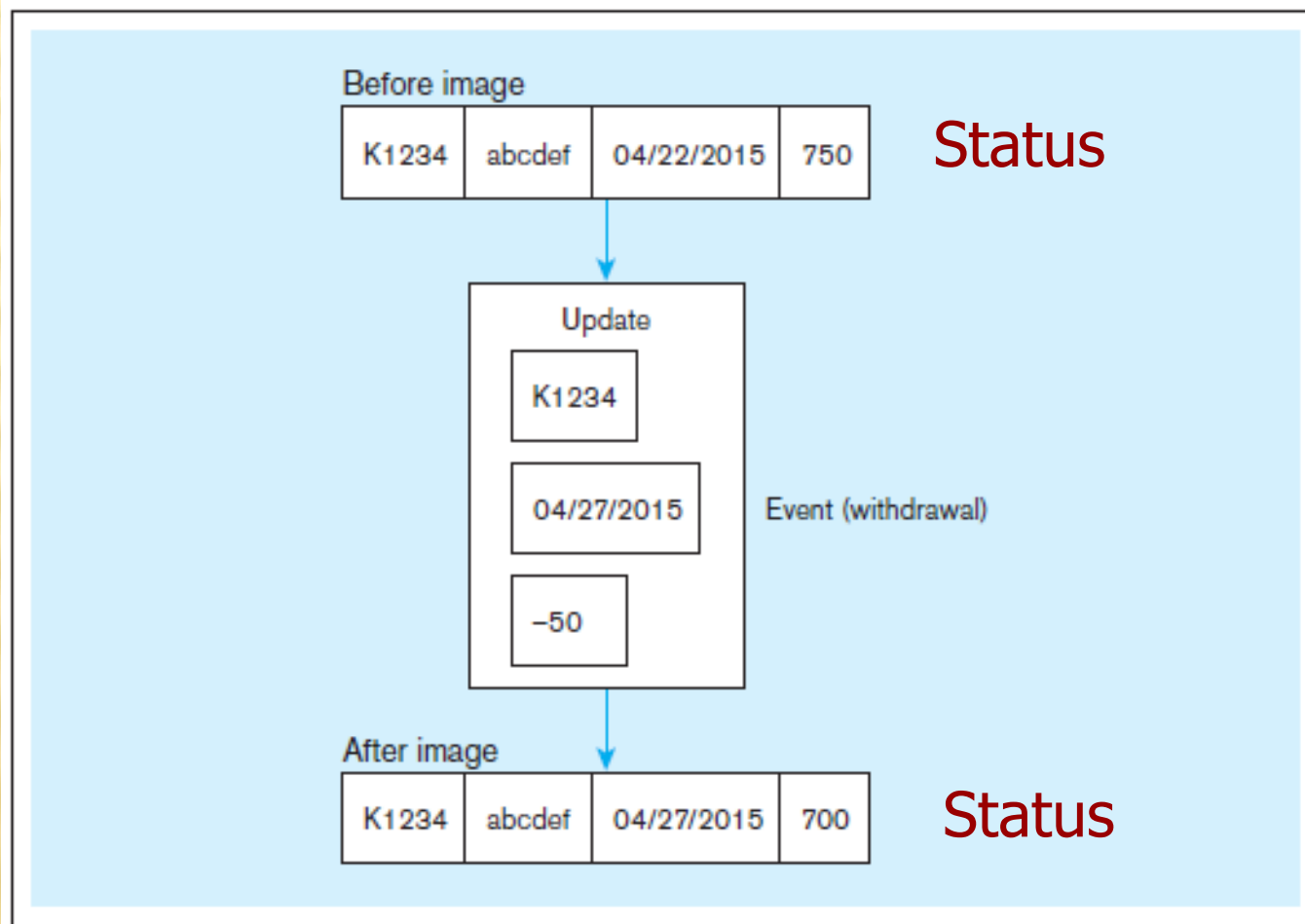


Figure 9-6
Example of DBMS
log entry

Event = a
database action
(create/ update/
delete) that
results from a
transaction

DATA CHARACTERISTICS

TRANSIENT VS. PERIODIC DATA

Table X (10/09)

Key	A	B
001	a	b
002	c	d
003	e	f
004	g	h

Table X (10/10)

Key	A	B
001	a	b
▶ 002	r	d
003	e	f
▶ 004	y	h
▶ 005	m	n

Table X (10/11)

Key	A	B
001	a	b
002	r	d
▶ 003	e	t
▶ []	[]	[]
005	m	n

Figure 9-7
Transient
operational data

With transient data, changes to existing records are written over previous records, thus destroying the previous data content.

DATA CHARACTERISTICS

TRANSIENT VS. PERIODIC DATA

Table X (10/09)

Key	Date	A	B	Action
001	10/09	a	b	C
002	10/09	c	d	C
003	10/09	e	f	C
004	10/09	g	h	C

Figure 9-8 Periodic warehouse data

Table X (10/10)

Key	Date	A	B	Action
001	10/09	a	b	C
002	10/09	c	d	C
▶ 002	10/10	r	d	U
003	10/09	e	f	C
004	10/09	g	h	C
▶ 004	10/10	y	h	U
▶ 005	10/10	m	n	C

Table X (10/11)

Key	Date	A	B	Action
001	10/09	a	b	C
002	10/09	c	d	C
002	10/10	r	d	U
003	10/09	e	f	C
▶ 003	10/11	e	t	U
004	10/09	g	h	C
004	10/10	y	h	U
▶ 004	10/11	y	h	D
005	10/10	m	n	C

Periodic data are never physically altered or deleted once they have been added to the store.

OTHER DATA WAREHOUSE CHANGES

- ✖ New descriptive attributes
- ✖ New business activity attributes
- ✖ New classes of descriptive attributes
- ✖ Descriptive attributes become more refined
- ✖ Descriptive data are related to one another
- ✖ New source of data

DERIVED DATA

× Objectives

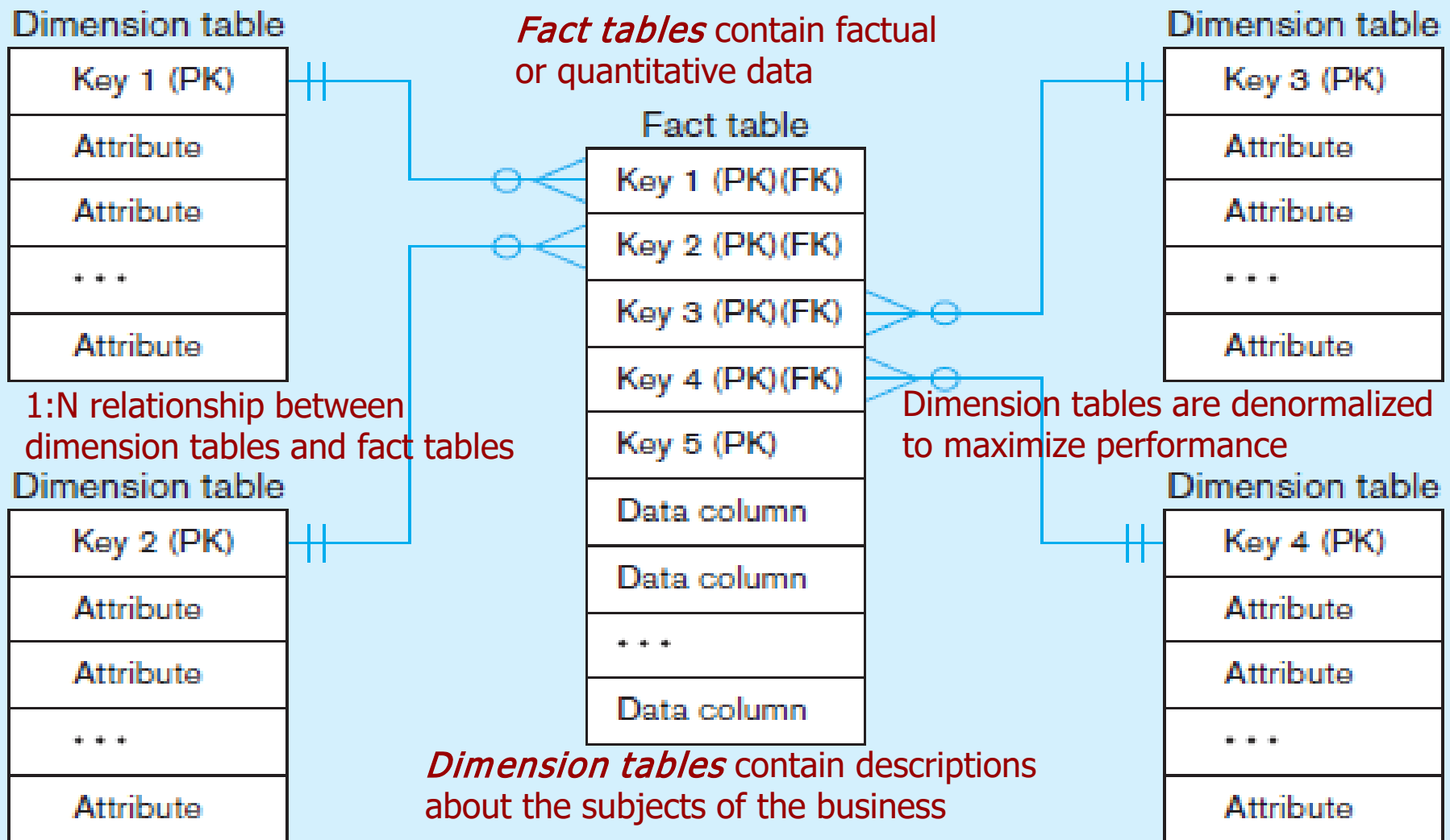
- + Ease of use for decision support applications
- + Fast response to predefined user queries
- + Customized data for particular target audiences
- + Ad-hoc query support
- + Data mining capabilities

× Characteristics

- + Detailed (mostly periodic) data
- + Aggregate (for summary)
- + Distributed (to departmental servers)

Most common data model = **dimensional model**
(usually implemented as a **star schema**)

Figure 9-9 Components of a **star schema**



Excellent for ad-hoc queries, but bad for online transaction processing

Figure 9-10 Star schema example

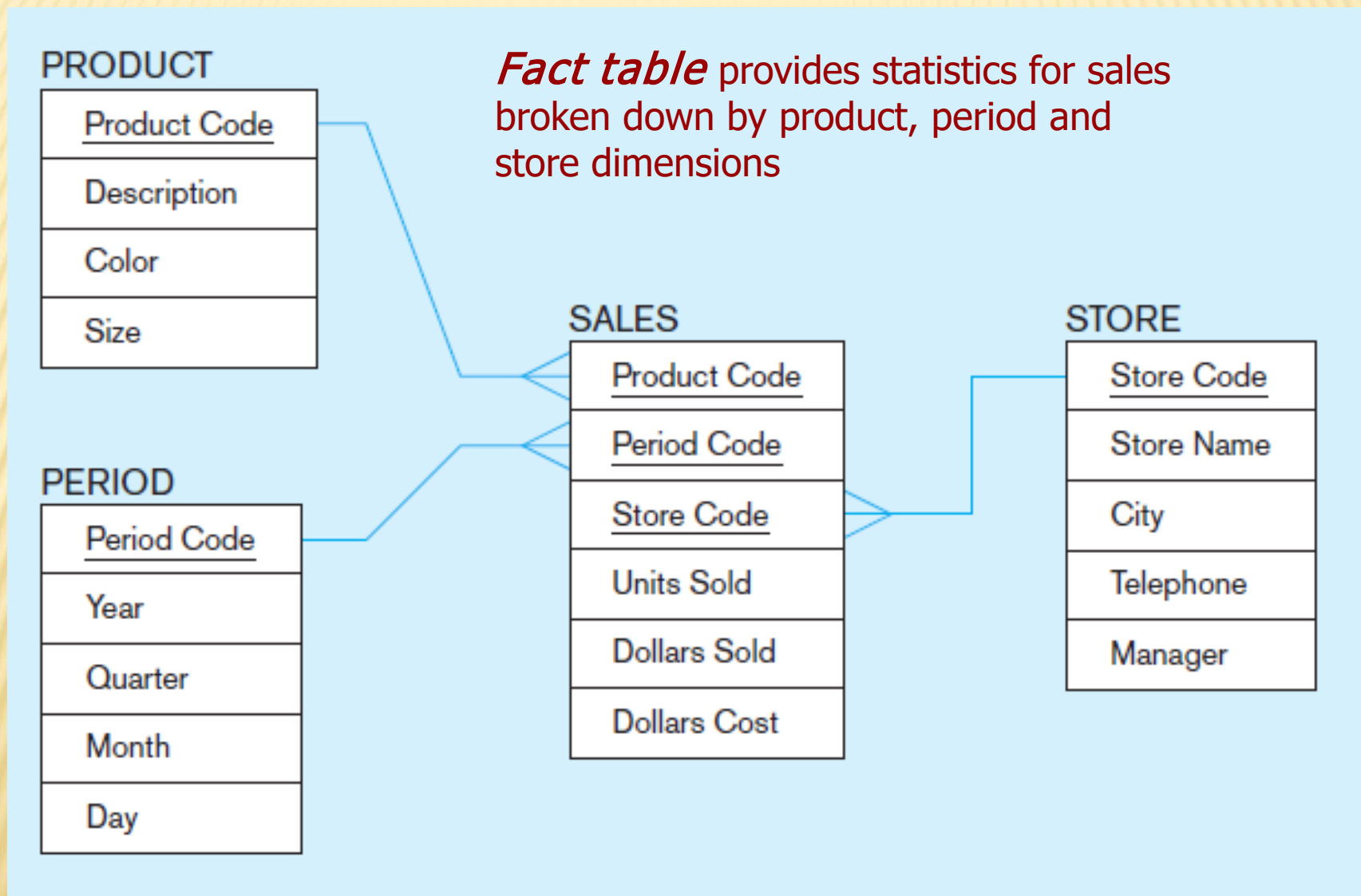
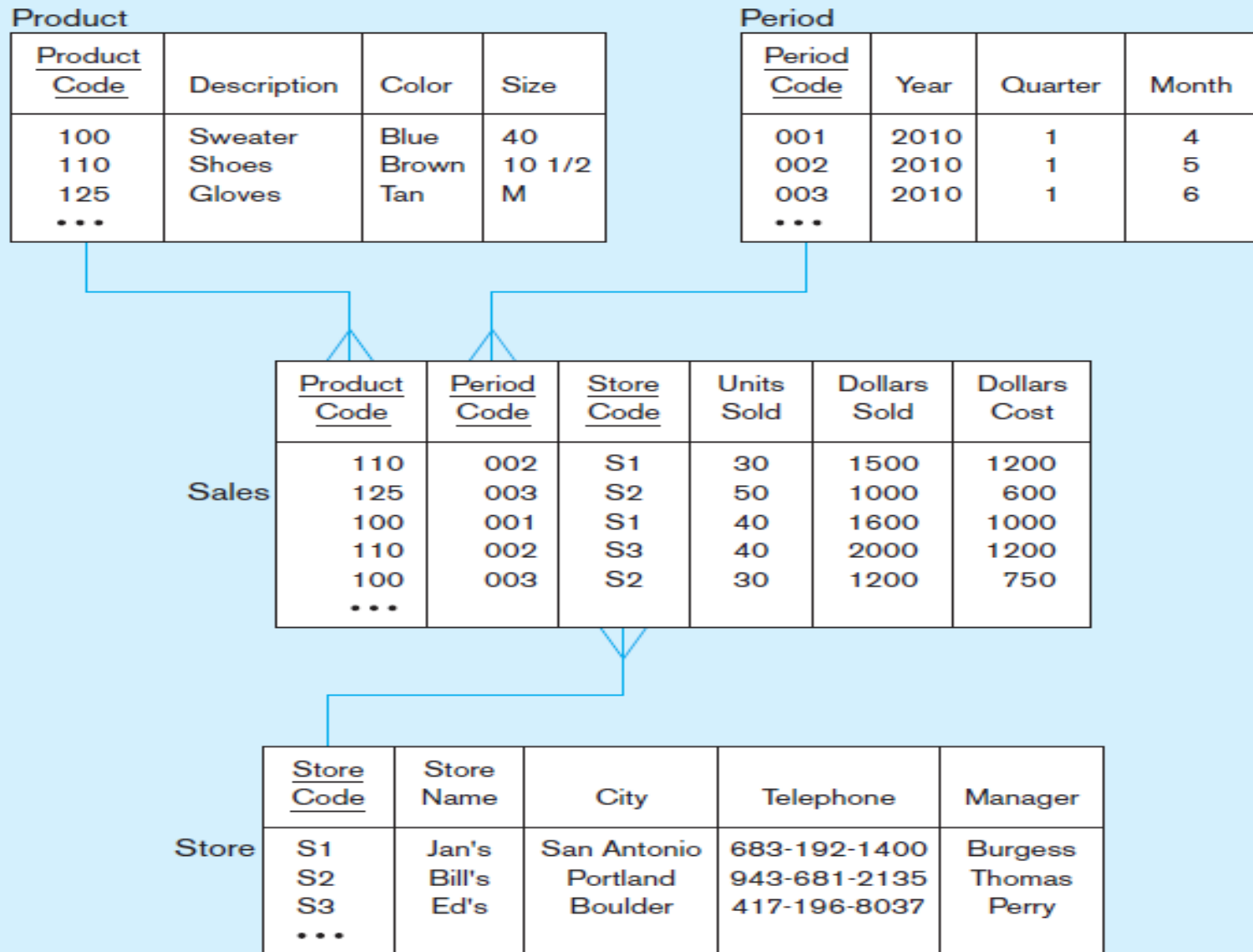


Figure 9-11 Star schema with sample data



SURROGATE KEYS

- ✖ Dimension table keys should be *surrogate* (non-intelligent and non-business related), because:
 - + Business keys may change over time
 - + Helps keep track of nonkey attribute values for a given production key
 - + Surrogate keys are simpler and shorter
 - + Surrogate keys can be same length and format for all key

GRAIN OF THE FACT TABLE

- ✗ Granularity of Fact Table–what level of detail do you want?
 - + Transactional grain–finest level
 - + Aggregated grain–more summarized
 - + Finer grains → better *market basket analysis* capability
 - + Finer grain → more dimension tables, more rows in fact table
 - + In Web-based commerce, finest granularity is a click

DURATION OF THE DATABASE

- + Natural duration–13 months or 5 quarters
- + Financial institutions may need longer duration
- + Older data is more difficult to source and cleanse

SIZE OF FACT TABLE

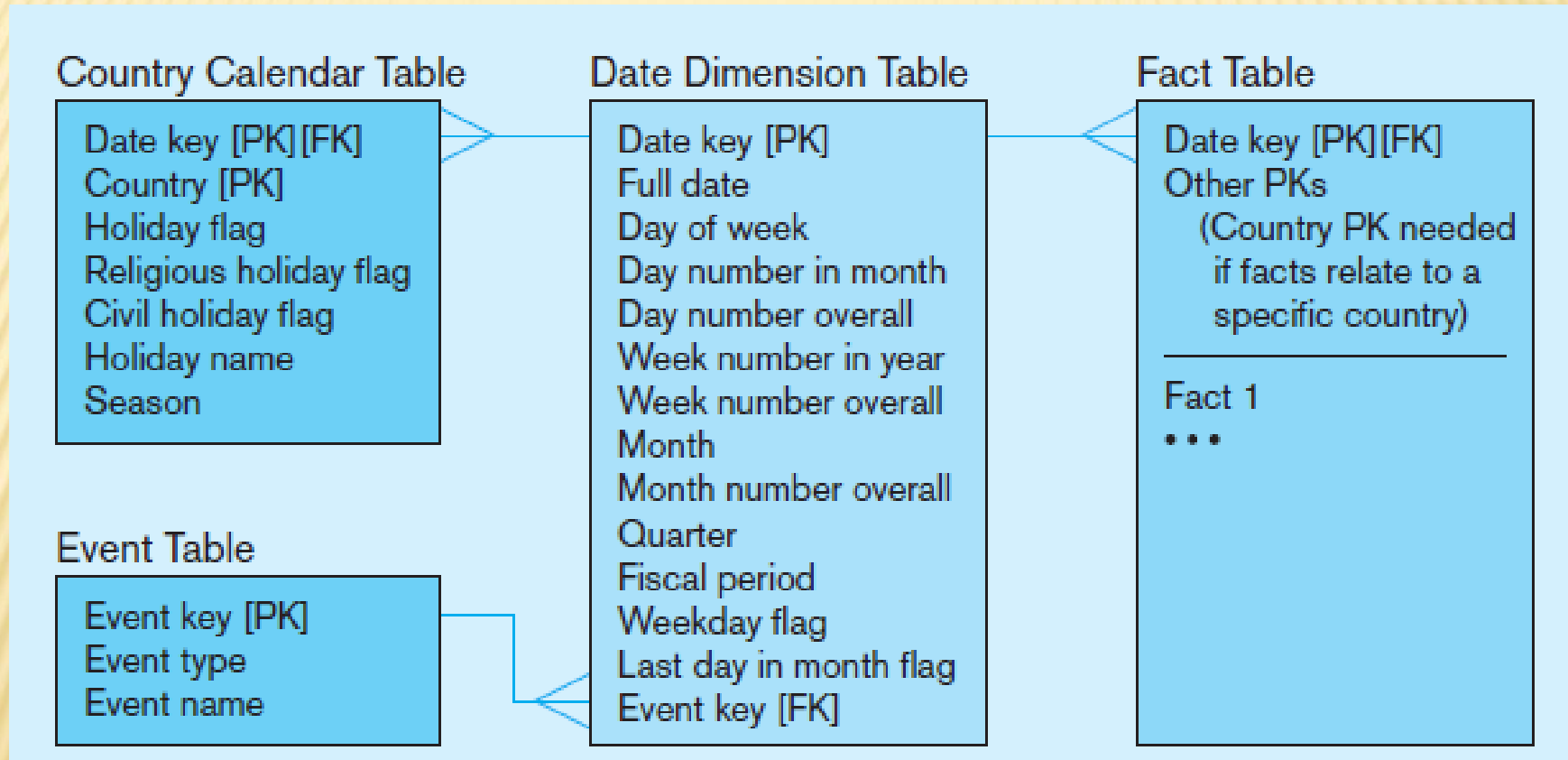
- ✗ Depends on the number of dimensions and the grain of the fact table
- ✗ Number of rows = product of number of possible values for each dimension associated with the fact table
- ✗ Example: Assume the following for Figure 9-11:

Total number of stores = 1,000
Total number of products = 10,000
Total number of periods = 24 (2 years' worth of monthly data)

- ✗ Total rows calculated as follows (assuming only half the products record sales for a given month):

Total rows = 1,000 stores × 5,000 active products × 24 months
= 120,000,000 rows (!)

Figure 9-12 Modeling dates



Fact tables contain time-period data
➔ Date dimensions are important

VARIATIONS OF THE STAR SCHEMA

✕ Multiple Facts Tables

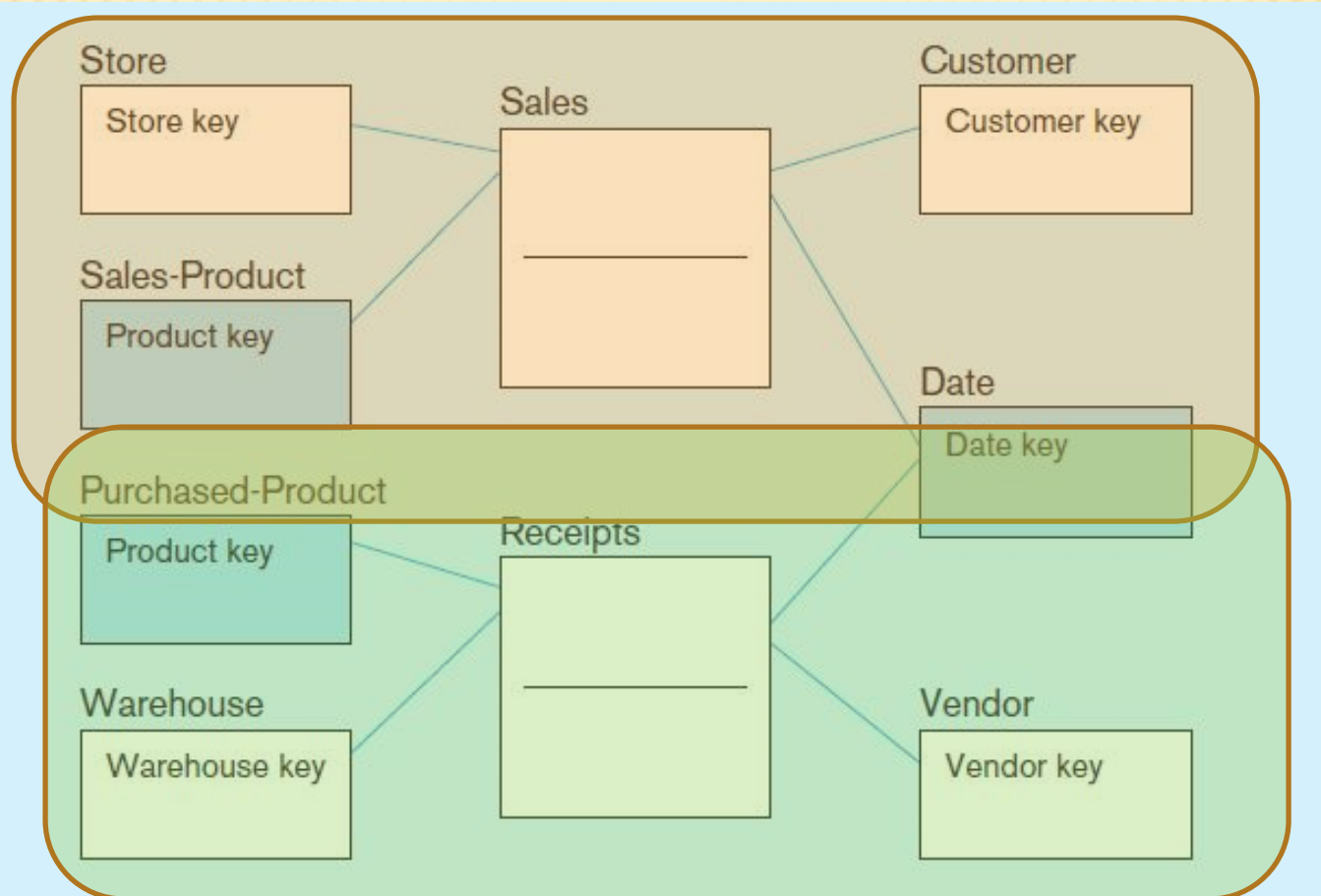
- + Can improve performance
- + Often used to store facts for different combinations of dimensions
- + Conformed dimensions

✕ Factless Facts Tables

- + No nonkey data, but foreign keys for associated dimensions
- + Used for:
 - ✕ Tracking events
 - ✕ Inventory coverage

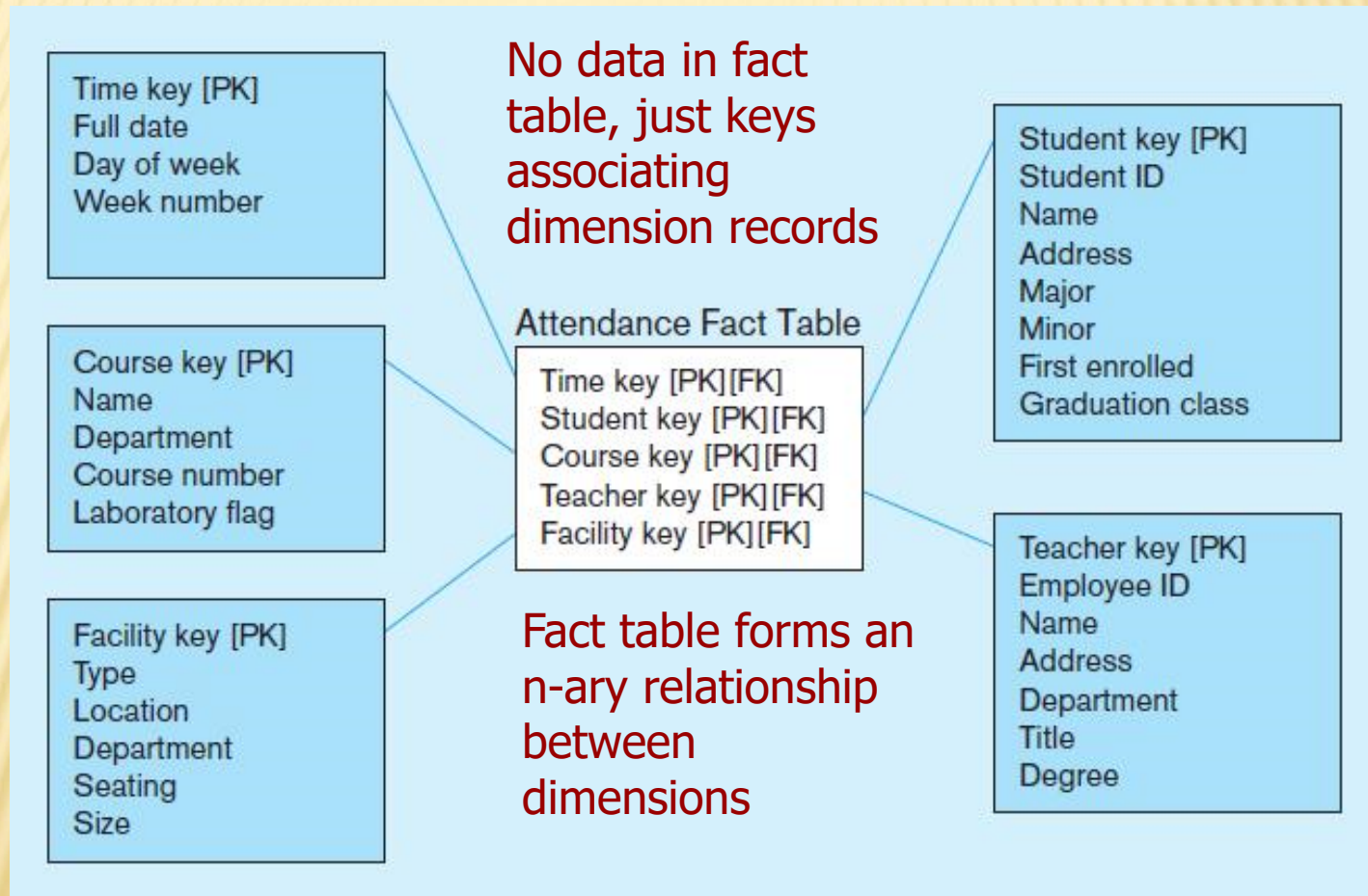
Figure 9-13 Conformed dimensions

Two fact tables → two (connected) star schemas.



Conformed dimension
Associated with multiple fact tables

Figure 9-14a Factless fact table showing occurrence of an event



NORMALIZING DIMENSION TABLES

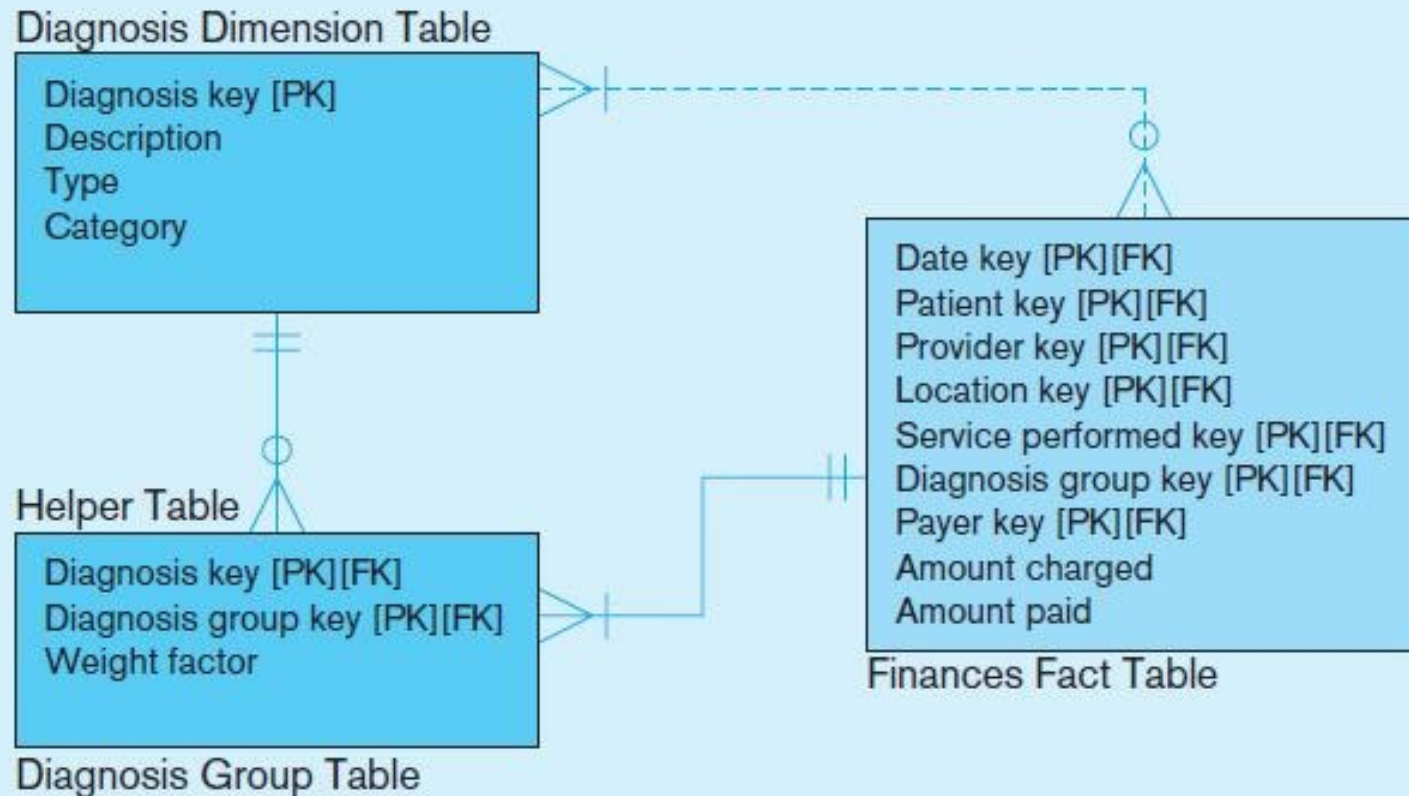
✖ Multivalued Dimensions

- + Facts qualified by a set of values for the same business subject
- + Normalization involves creating a table for an associative entity between dimensions

✖ Hierarchies

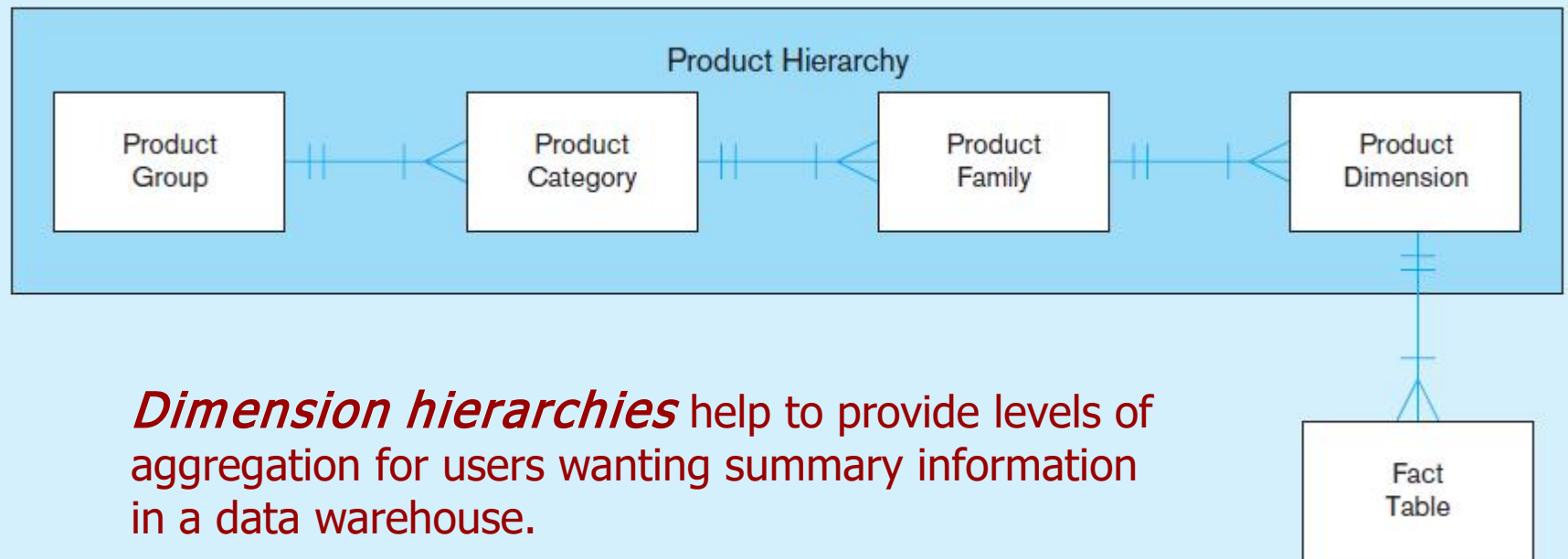
- + Sometimes a dimension forms a natural, fixed depth hierarchy
- + Design options
 - ✖ Include all information for each level in a single denormalized table
 - ✖ Normalize the dimension into a nested set of 1:M table relationships

Figure 9-15 Multivalued dimension



Helper table is an associative entity that implements a M:N relationship between dimension and fact.

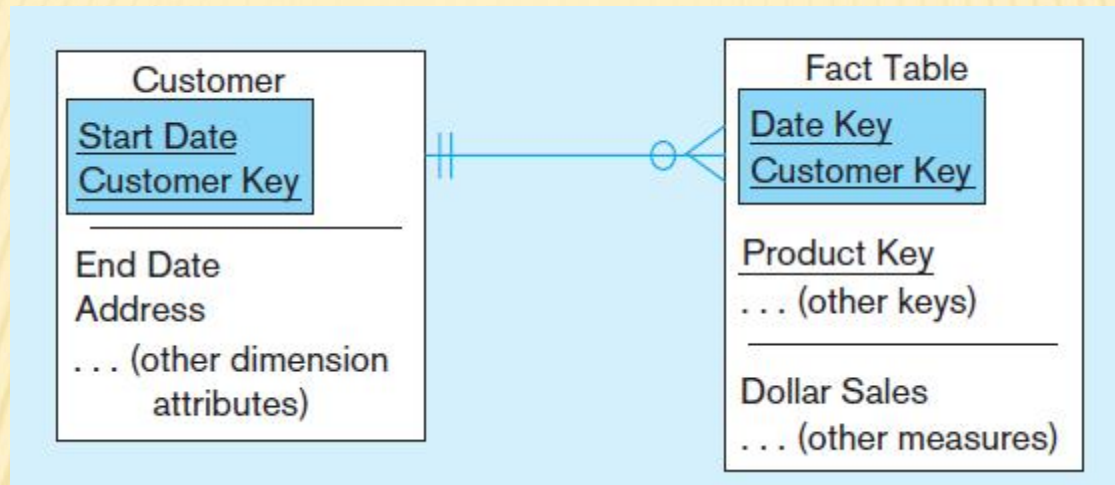
Figure 9-16 Fixed product hierarchy



SLOWLY CHANGING DIMENSIONS (SCD)

- ✖ How to maintain knowledge of the past
- ✖ Kimball's approaches:
 - + Type 1: just replace old data with new (lose historical data)
 - + Type 2: for each changing attribute, create a current value field and several old-valued fields (multivalued)
 - + Type 3: create a new dimension table row each time the dimension object changes, with all dimension characteristics at the time of change. Most common approach

Figure 9-18 Example of Type 2 SCD Customer dimension table

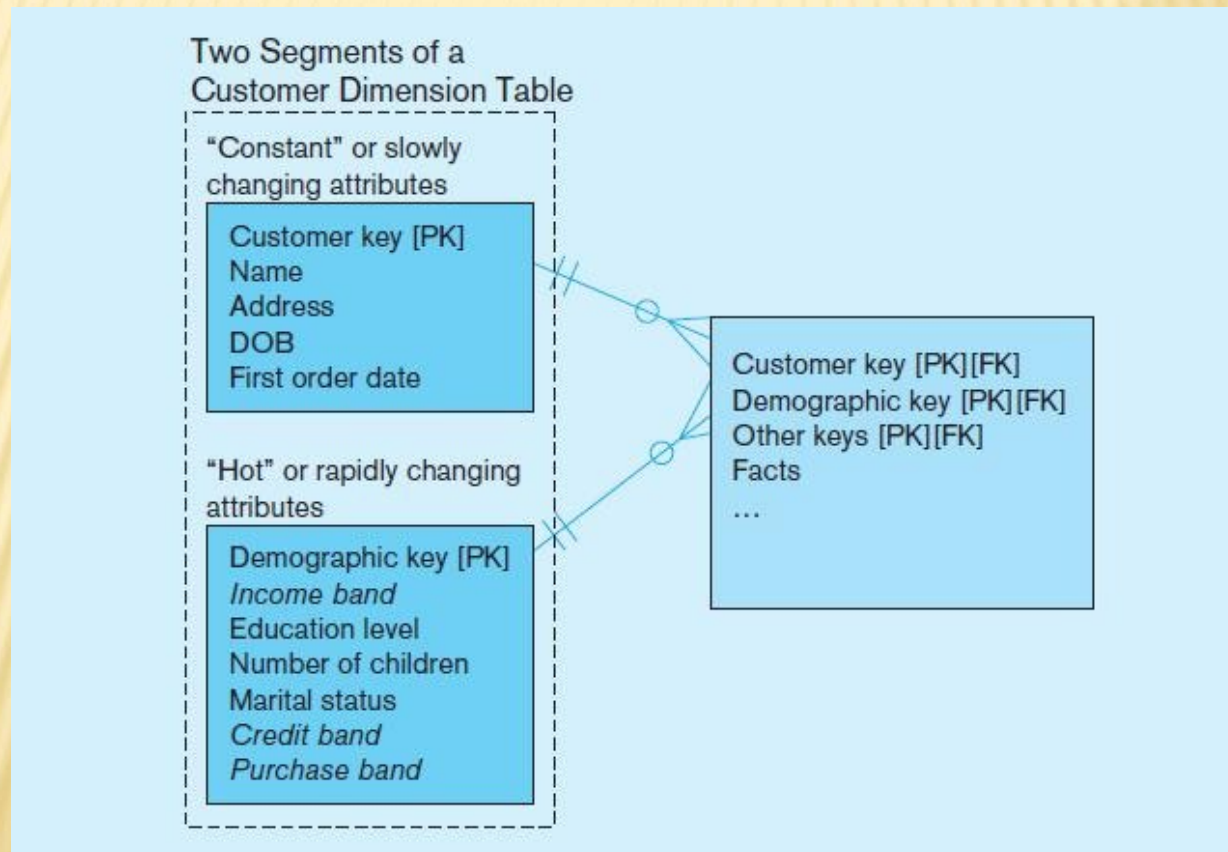


The dimension table contains several records for the same customer. The specific customer record to use depends on the key and the date of the fact, which should be between start and end dates of the SCD customer record.

```
WHERE Fact.CustomerKey = Customer.CustomerKey  
AND Fact.DateKey BETWEEN Customer.StartDate and Customer.EndDate
```

Figure 9-19 Dimension segmentation

For rapidly changing attributes (hot attributes), Type 2 SCD approach creates too many rows and too much redundant data. Use segmentation instead.



10 ESSENTIAL RULES FOR DIMENSIONAL MODELING

- ✗ Use atomic facts
- ✗ Create single-process fact tables
- ✗ Include a date dimension for each fact table
- ✗ Enforce consistent grain
- ✗ Disallow null keys in fact tables
- ✗ Honor hierarchies
- ✗ Decode dimension tables
- ✗ Use surrogate keys
- ✗ Conform dimensions
- ✗ Balance requirements with actual data

THE FUTURE OF DATA WAREHOUSING: INTEGRATION WITH BIG DATA AND ANALYTICS

- ✗ Issue of Big Data (huge volume, often unstructured)
- ✗ Speed of processing
 - ✗ Design/purchase storage, database, and networking aspects in tandem
 - ✗ Use in-memory databases (RAM instead of disk)
 - ✗ Add analytics capabilities closer to the original data sources instead of separate data warehouses
- ✗ Cost of Data Storage
 - + Move data warehouse to the cloud
 - + Columnar databases optimize storage
- ✗ Unstructured Data
 - + NoSql “Not only SQL”
 - + Hadoop